

Part 1: optimization

Review: classical optimization theory

A modern take: acceleration via large stepsizes

Summary, open problems, Q&A

Part 2: generalization

Review: descent lemma

For GD, $L(\theta_t)$ decreases **monotonically** for **small η** such that

$$\eta < \frac{2}{\sup \|\nabla^2 L(\cdot)\|}$$

Proof.

$$L(\theta_+) = L(\theta - \eta \nabla L(\theta))$$

GD step

$$= L(\theta) - \eta \|\nabla L(\theta)\|^2 + \frac{\eta^2}{2} \nabla L(\theta)^\top \nabla^2 L(\nu) \nabla L(\theta)$$

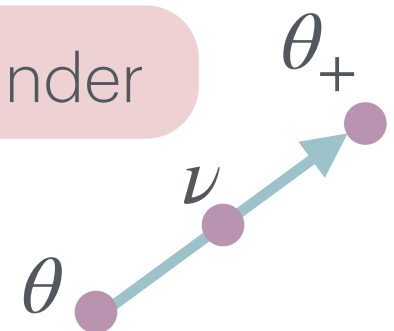
Taylor remainder

$$\leq L(\theta) - \eta \|\nabla L(\theta)\|^2 \left(1 - \frac{\eta}{2} \|\nabla^2 L(\nu)\| \right)$$

operator norm

$$\leq L(\theta)$$

small stepsize



this descent lemma can be generalized

Review: convergence rates

Let L be 1-smooth ($\|\nabla^2 L\| \leq 1$) with finite minimizer w^* . For GD with $\eta = 1$, we have

descent lemma

$$L(\theta_t) \downarrow$$

convexity

$$L(\theta_t) - \min L \leq \frac{\|\theta_0 - \theta^*\|^2}{2t}$$

α -strong convexity

$$L(\theta_t) - \min L \leq e^{-\alpha t} (L(\theta_0) - \min L)$$

number of steps to get ϵ -error:

$$O(1/\epsilon) \text{ and } O(\kappa \log(1/\epsilon))$$

$\kappa = 1/\alpha$, condition number

Review: gradient flow analysis

For convex L and gradient flow $d\theta_t = -\nabla L(\theta_t)dt$, we have

$$L(\theta_t) - L(\nu) \leq \frac{\|\theta_0 - \nu\|^2}{2t} \quad \text{for all } \nu$$

Proof.

step 1:

chain rule

gradient flow

convexity

$$d\frac{1}{2}\|\theta_t - \nu\|^2 = \langle \theta_t - \nu, d\theta_t \rangle = \langle \theta_t - \nu, -\nabla L(\theta_t) \rangle dt \leq L(\nu) - L(\theta_t)$$

step 2:

integration

descent lemma

$$\frac{1}{2}\|\theta_t - \nu\|^2 - \frac{1}{2}\|\theta_0 - \nu\|^2 \leq \int_0^t L(\nu) - L(\theta_s) ds \leq t(L(\nu) - L(\theta_t))$$

step 3: rearranging terms

for small stepsize, discretize this => GD analysis

Review: acceleration

number of steps to get ϵ -error

GD

$$\theta_+ = \theta - \eta \nabla L(\theta)$$

$$O(1/\epsilon) \text{ \& } O(\kappa \log(1/\epsilon))$$

Nesterov's momentum

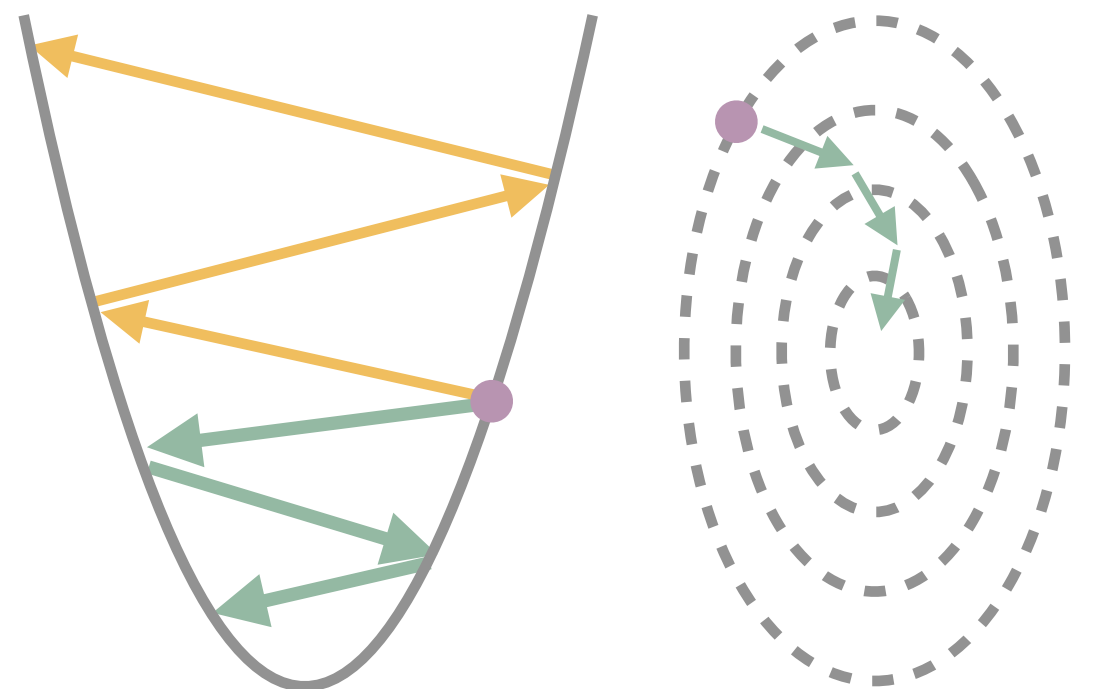
$$\theta_+ = \nu - \eta \nabla L(\nu)$$

$$\nu_+ = \theta_+ + \beta(\theta_+ - \theta)$$

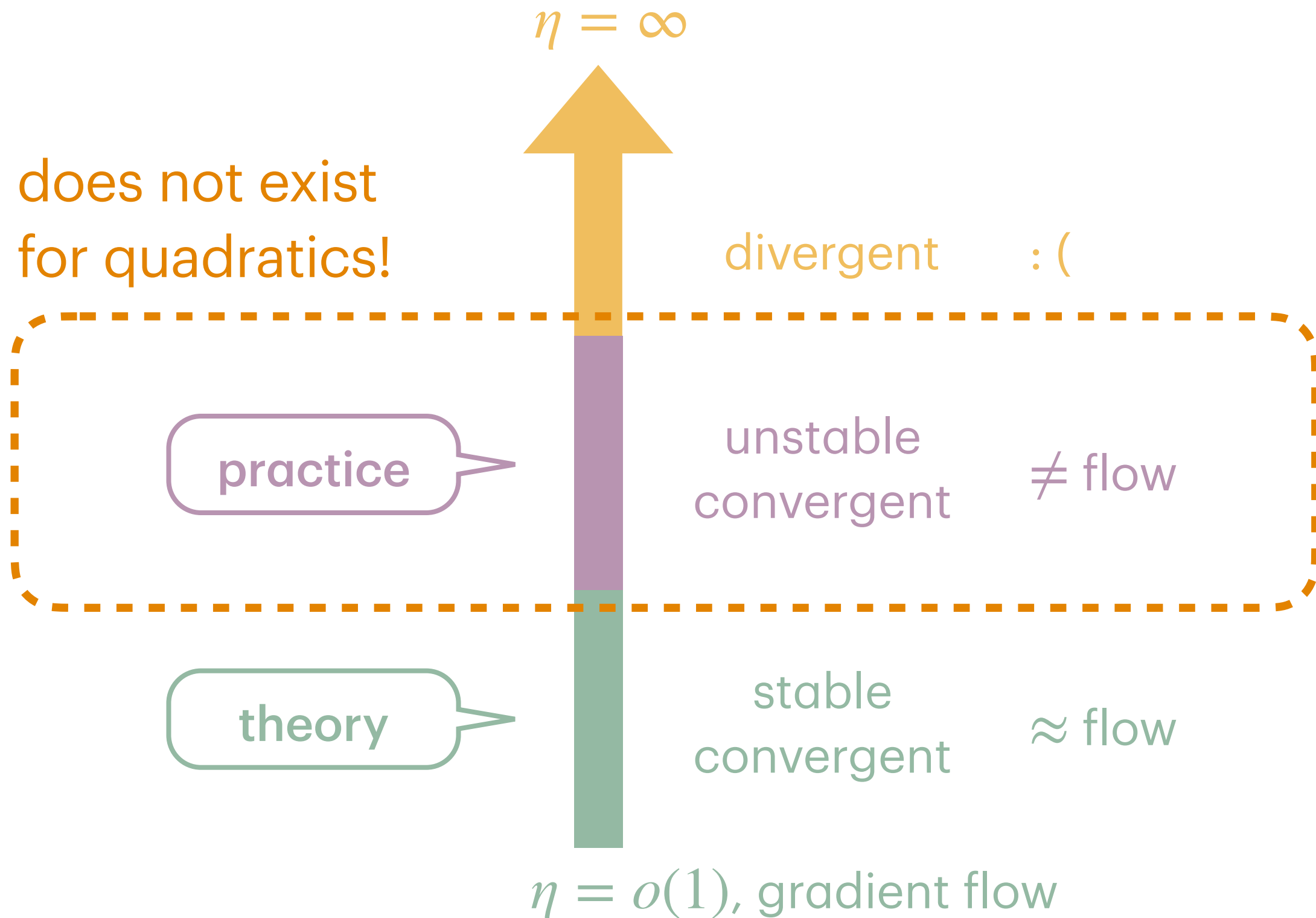
$$O(1/\sqrt{\epsilon}) \text{ \& } O(\sqrt{\kappa} \log(1/\epsilon))$$

these rates are optimal

hard case: quadratics in high-dim



From small to large stepsize



Alternative mental model

linear
regression

logistic
regression

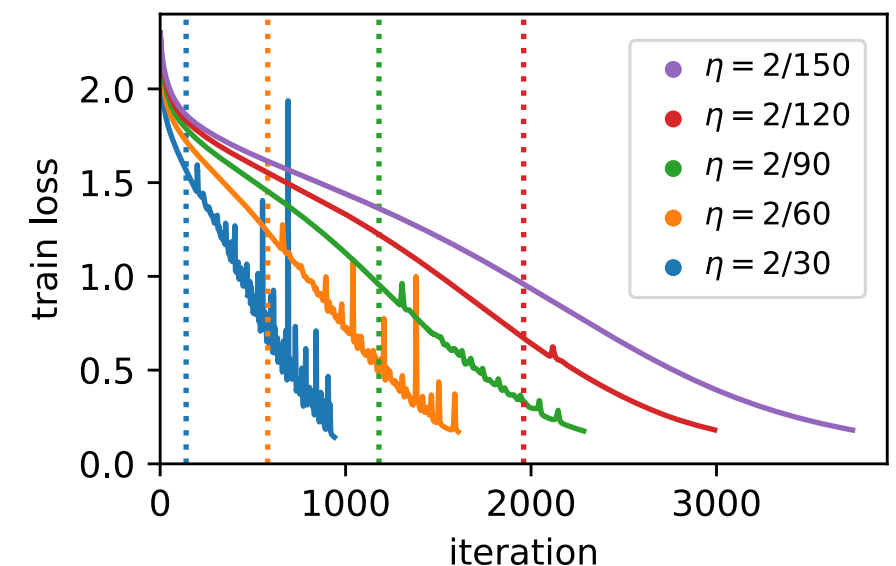
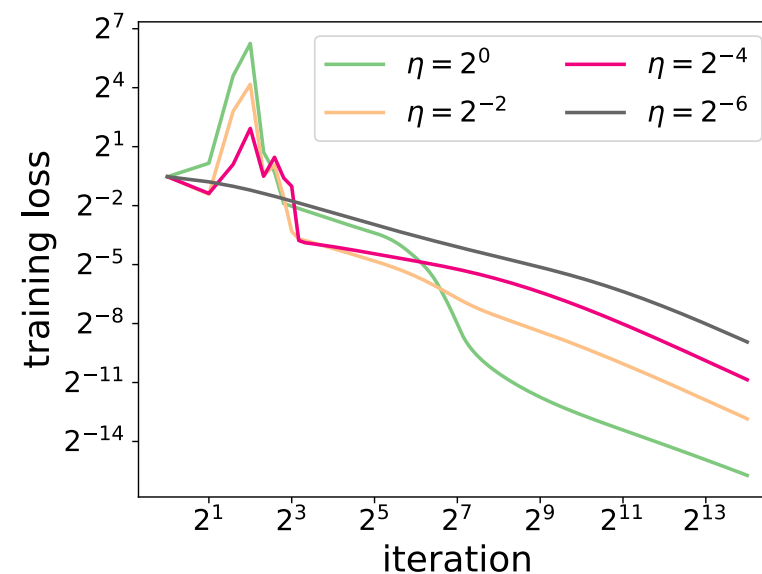
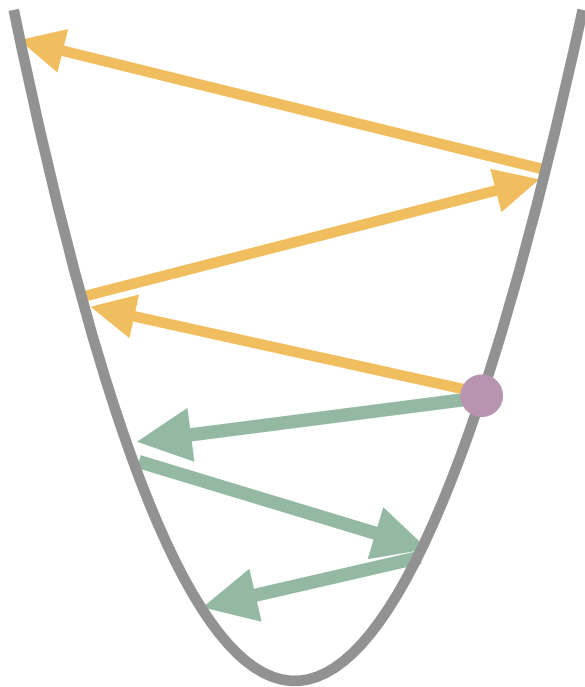
.....

deep
learning

unstable
convergence
impossible

observable
& provable

unstable
convergence
observed



(1/3) Logistic regression

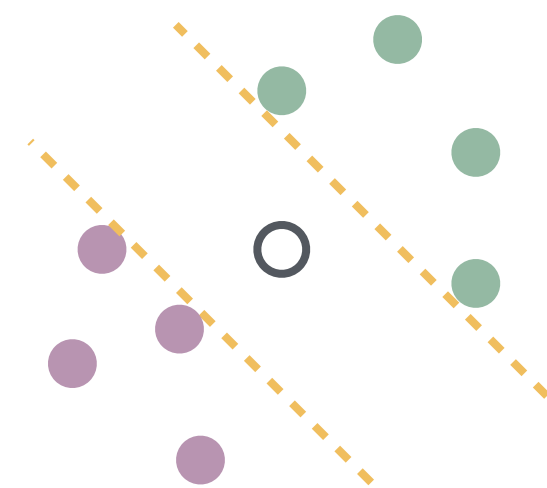
$$L(\theta) = \frac{1}{n} \sum_{i=1}^n \ln(1 + \exp(-y_i x_i^\top \theta))$$

smooth, convex
non-strongly convex

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

Assumption (bounded + separable)

- $\|x_i\| \leq 1, y_i \in \{\pm 1\}, i = 1, \dots, n$
- $\exists$ unit vector $\theta^*, \min_i y_i x_i^\top \theta^* \geq \gamma > 0$



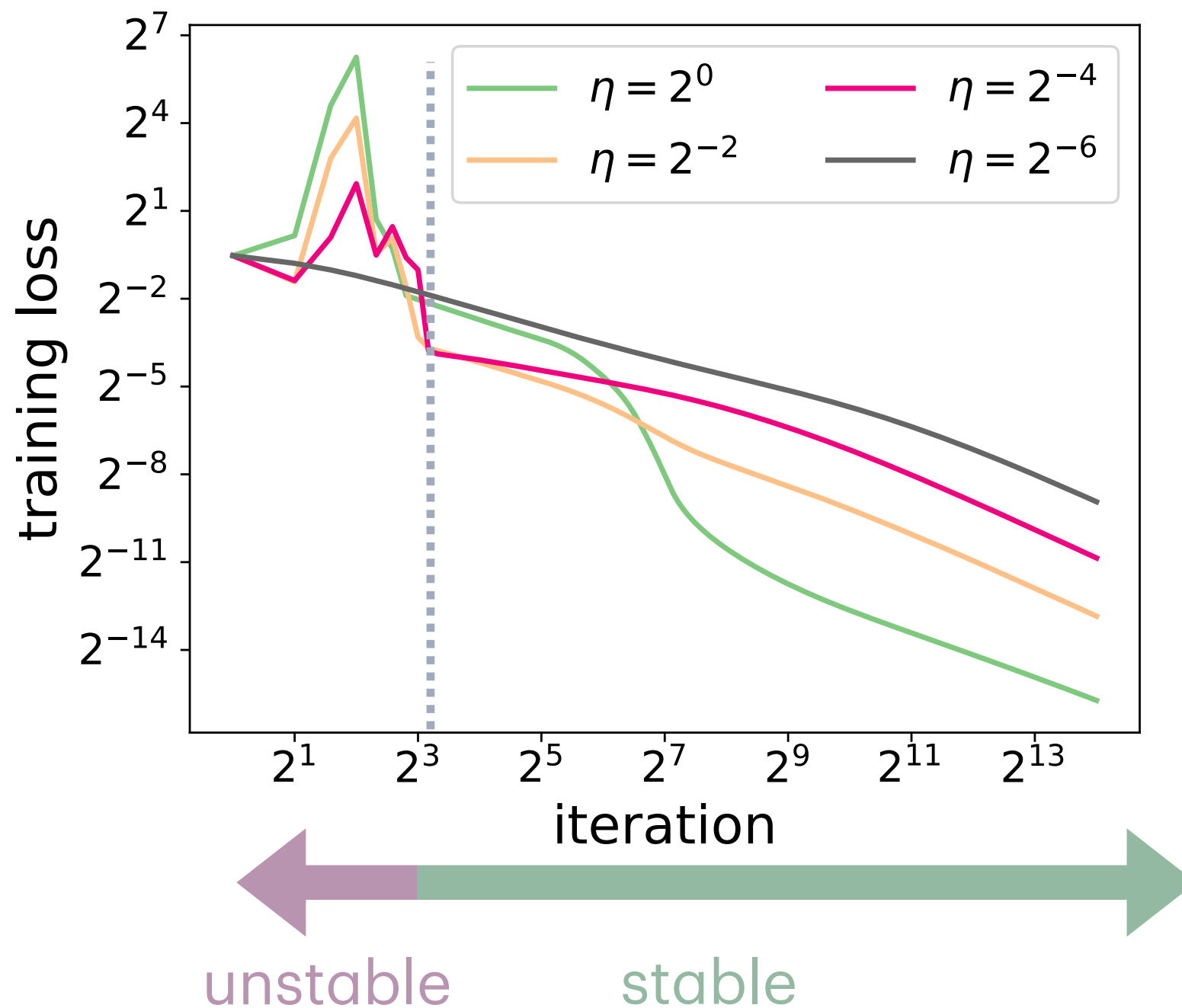
Classical theory

“almost surely” if overparameterized

For $\eta = \Theta(1)$, $L(\theta_t) \downarrow$ and $L(\theta_t) = \tilde{O}(1/t)$

improved to $\tilde{O}(1/t^2)$ by Nesterov

(1/3) Logistic regression



s -th step is in **stable phase** if $L(\theta_t) \downarrow$ for all $t \geq s$
unstable phase if otherwise

(1/3) Theorem

Unstable phase.

for any η and t ,

$$\frac{1}{t} \sum_{k=0}^{t-1} L(\theta_k) = \tilde{O}\left(\frac{1 + \eta^2}{\eta t}\right)$$

tendency to decrease

Phase transition.

GD exits unstable phase in τ steps for

$$\tau = \Theta(\eta)$$

$$\tau = \Theta\left(\max\{\eta, n, n/\eta \ln(n/\eta)\}\right)$$

Stable phase.

$$L(\theta_{\tau+t}) \downarrow \text{ and } L(\theta_{\tau+t}) = \tilde{O}\left(\frac{1}{\eta t}\right)$$

“flow rate”

(1/3) Effects of large stepsize

1. Asymptotic $1/(\eta t)$ rate $\Rightarrow$ 2x stepsize 2x faster
2. Phase transition in $\Theta(\eta)$ steps $\Rightarrow$ longer unstable phase
3. Given #steps $T \geq \Theta(n)$, if choose $\eta = \Theta(T)$, then

$$\tau \leq T/2 \text{ and } L(\theta_T) = \tilde{O}(1/T^2)$$

A lower bound. There exists a separable dataset, if η is such that $L(\theta_t) \downarrow$ for all t , then

$$L(\theta_t) = \Omega(1/t)$$

acceleration by large stepsize

Wu, Bartlett, Telgarsky, Yu. “Large stepsize gradient descent for logistic loss: non-monotonicity of the loss improves optimization efficiency.” COLT 2024

(1/3) A “non-quadratic” picture

$$\exists \text{ unit vector } \theta^*, \min_i y_i x_i^\top \theta^* > \gamma > 0$$

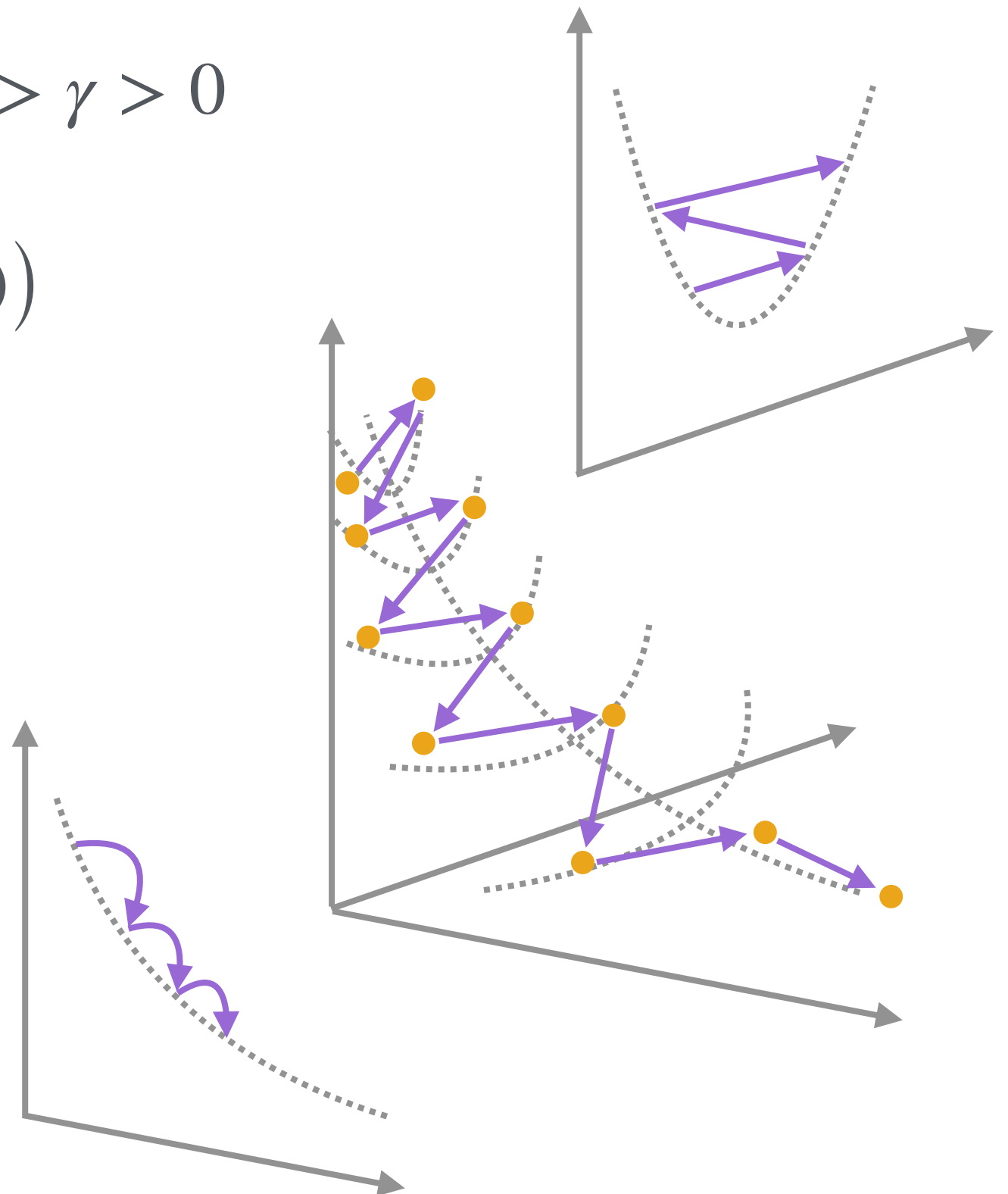
$$L(\theta) = \hat{\mathbb{E}} \ln(1 + \exp(-y x^\top \theta))$$

minimizer at ∞

$$\lim_{\lambda \rightarrow \infty} L(\lambda \theta^*) = 0$$

self-bounded

$$\|\nabla^2 L\| \leq L$$



Two extensions

minimizer at ∞

$$\lim_{\lambda \rightarrow \infty} L(\lambda \theta^*) = 0$$

finite minimizer

e.g. regularization

*unstable
convergence under
finite minimizer*

self-bounded

$$\|\nabla^2 L\| \leq L$$

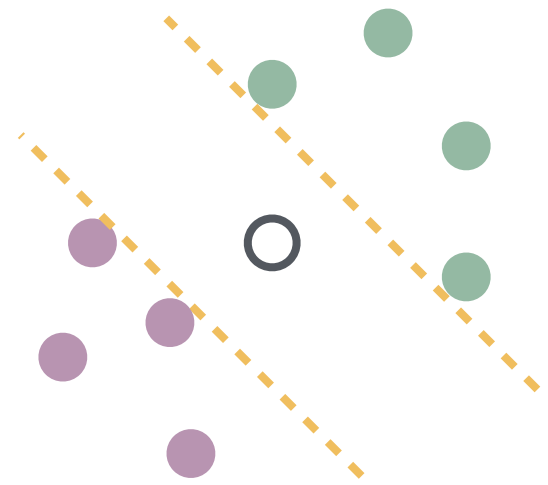
enabling “tricks”

e.g. adaptive GD
[Ji & Telgarsky 2021]

*large stepsizes for
GD variants*

(2/3) Large, adaptive stepsize

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i x_i^\top \theta) \quad \ell(t) = \ln(1 + \exp(-t))$$



$$\theta_{t+1} = \theta_t - \eta \left((-\ell^{-1})' \circ L(\theta_t) \right) \nabla L(\theta_t)$$

adapt to curvature

$$\approx \theta_t - \frac{\eta}{L(\theta_t)} \nabla L(\theta_t)$$

↙

$$\theta_{t+1} = \theta_t - \eta \nabla \phi(\theta_t)$$

$$\phi(\theta) = -\ell^{-1}(L(\theta))$$
$$\approx \ln \sum \exp(-y_i x_i^\top \theta)$$

[Ji & Telgarsky, 2021]

For $\eta = \Theta(1)$, $L(\theta_t) \downarrow$ and $L(\theta_t) \leq \exp(-\Theta(t))$

large stepsize makes adaptive GD even faster

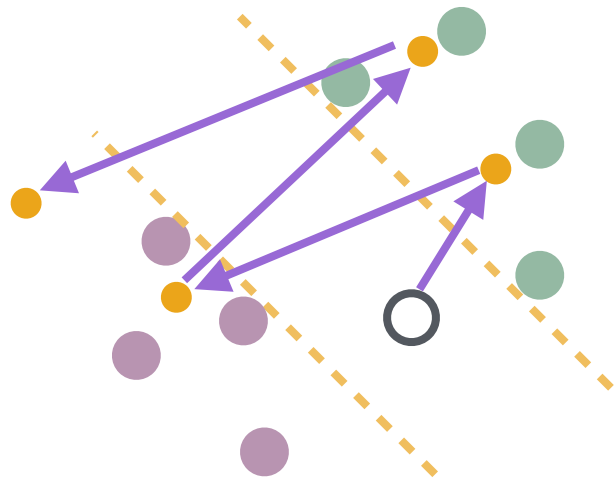
(2/3) Theorem

Assume separability with margin γ . For $t \geq 1/\gamma^2$ and every η

$$L(\bar{\theta}_t) \leq \exp(-\Theta(\gamma^2 \eta t)), \quad \text{where } \bar{\theta}_t = \frac{1}{t} \sum_{k=1}^t \theta_k$$

arbitrarily small error in $1/\gamma^2$ steps

$$\lim_{\eta \rightarrow \infty} L(\bar{\theta}_t) = 0 \quad \text{for } t = 1/\gamma^2$$



matching “Perceptron”
[Novikoff, 1962, or earlier]

(2/3) Theorem (lower bound)

$\forall \theta_0, \exists (x_i, y_i)_{i=1}^n$ with margin γ such that: for any first-order batch method

$$\min_i y_i x_i^\top \theta_t > 0 \Rightarrow t \geq \Omega(1/\gamma^2)$$

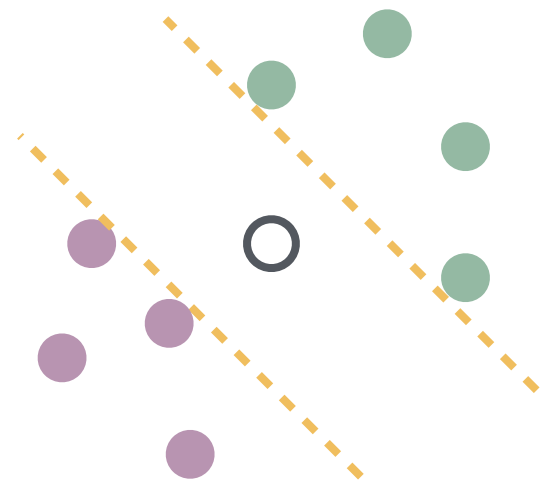
first-order batch method:

$$\theta_t \in \theta_0 + \text{span}\{ \nabla L(\theta_0), \dots, \nabla L(\theta_{t-1}) \}$$

where $L(\theta) = \hat{\mathbb{E}} \ell(yx^\top \theta)$ for any ℓ

large, adaptive stepsizes = minimax optimal

(3/3) ℓ_2 -regularization



$\Theta(1)$ -smooth, λ -strongly convex
condition number $\kappa = \Theta(1/\lambda)$

$$\tilde{L}(\theta) = L(\theta) + \frac{\lambda}{2} \|\theta\|^2 \quad L(\theta) = \frac{1}{n} \sum_{i=1}^n \ln(1 + \exp(-y_i x_i^\top \theta))$$

finite minimizer $\tilde{\theta}$ with norm $\|\tilde{\theta}\| = O(\ln \kappa)$

$$\text{GD} \quad \theta_{t+1} = \theta_t - \eta \nabla \tilde{L}(\theta_t)$$

Classical theory

For $\eta = \Theta(1)$, $\tilde{L}(\theta_t) \downarrow$ and $\tilde{L}(\theta_t) - \min \tilde{L} \leq \epsilon$ for $t = O(\kappa \ln(1/\epsilon))$

improved to $\tilde{O}(\sqrt{\kappa})$ by Nesterov

(3/3) Theorem

for $\lambda \leq \Theta(1)$, improvement is $\tilde{O}(\kappa^{2/3})$

Let $\kappa = 1/\lambda$. Assume separability and

$$\eta_{\max} = \Theta(\sqrt{\kappa})$$

$$\lambda \leq \Theta\left(\frac{1}{n \ln n}\right) \quad \eta \leq \Theta(\min\{\sqrt{\kappa}, \kappa/n\})$$

Phase transition. GD exits unstable phase in τ steps for

$$\tau := \Theta(\max\{\eta, n, n/\eta \ln(n/\eta)\})$$

$$\tau = \Theta(\sqrt{\kappa})$$

Stable phase. $\tilde{L}(\theta_{\tau+t}) \downarrow$ and

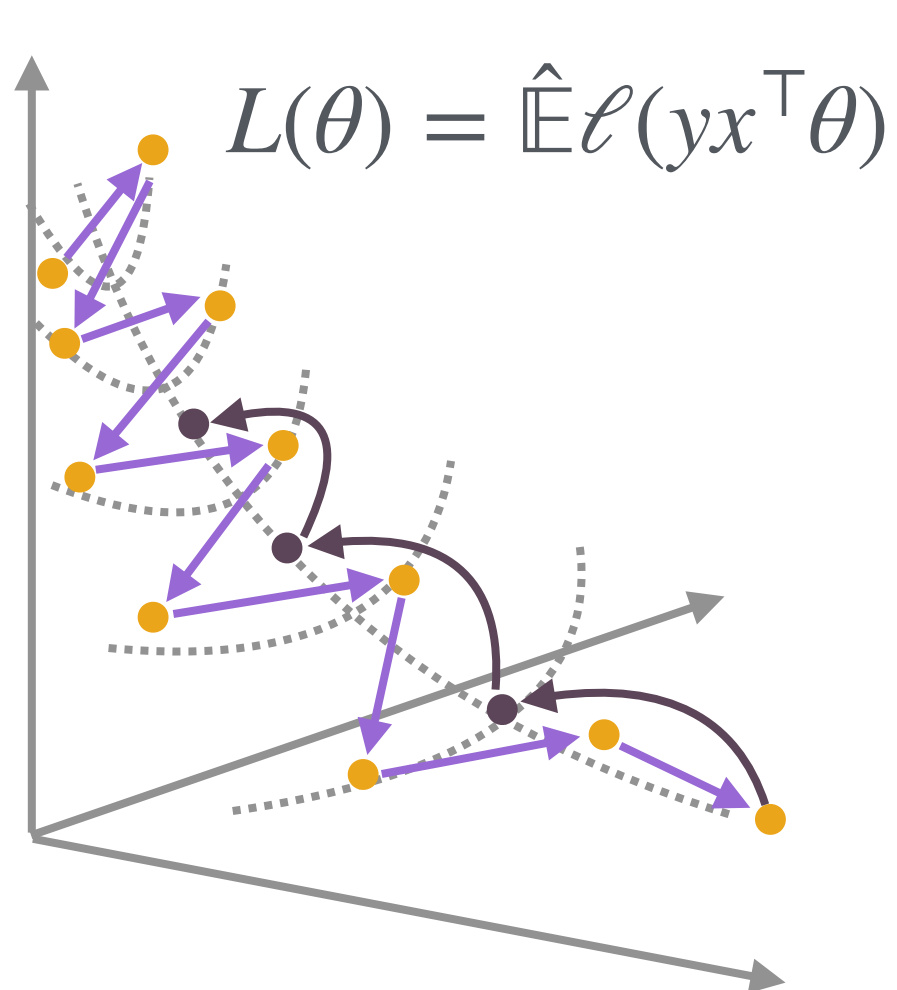
$$t = \Theta(\sqrt{\kappa} \ln(1/\epsilon))$$

$$\tilde{L}(\theta_{\tau+t}) - \min \tilde{L} \lesssim \exp(-t\eta/\kappa)$$

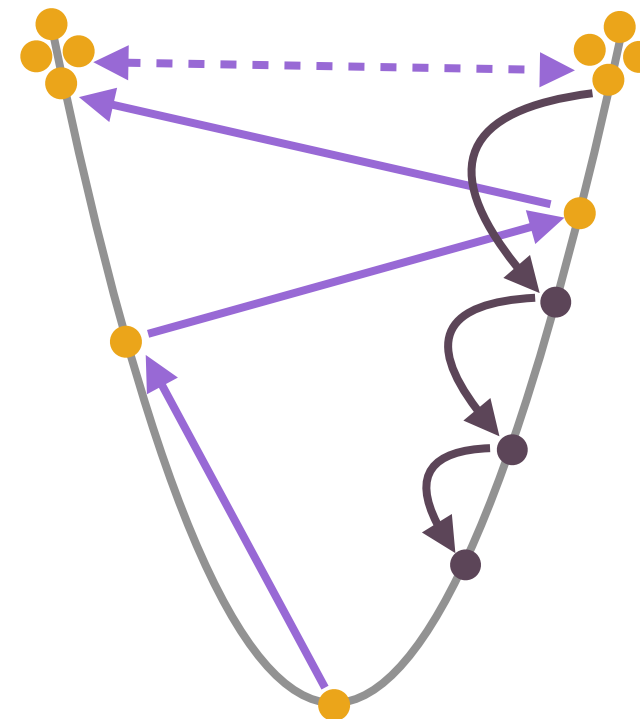
from $\tilde{O}(\kappa)$ to $\tilde{O}(\sqrt{\kappa})$: acceleration via large stepsize

Wu, Marion, Bartlett. "Large stepsizes accelerate gradient descent for regularized logistic regression." NeurIPS 2025

(3/3) Picture: valley + basin



$$R(\theta) = \frac{\lambda}{2} \|\theta\|^2$$



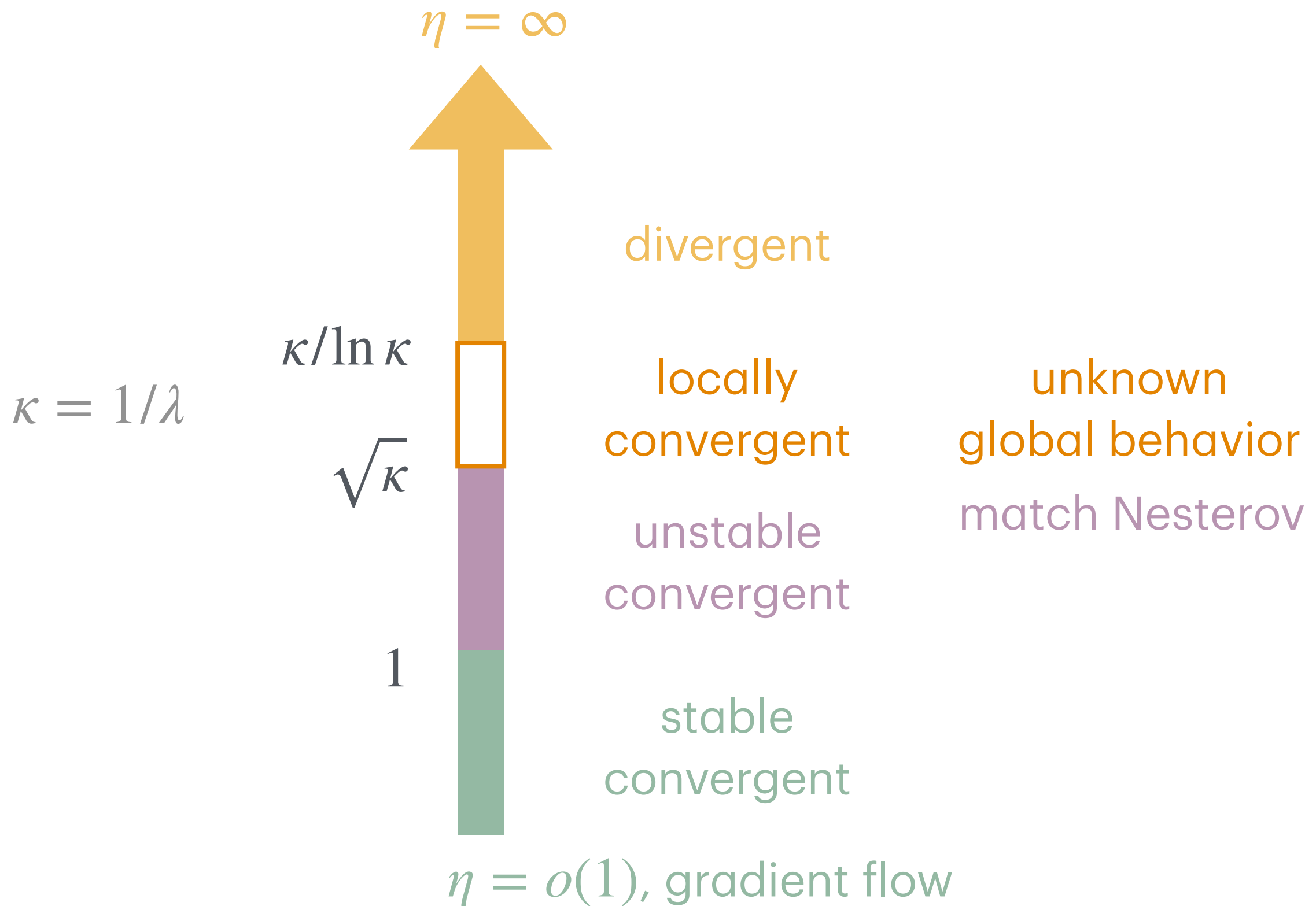
Unstable. $\tilde{L} \approx L, R \leq \Theta(1)$, “overshoot”

$$\|\tilde{\theta}\| = O(\ln \kappa)$$

Stable. “move back”

$$\sup \|\theta_t\| = \Theta(\eta) = \text{poly}(\kappa)$$

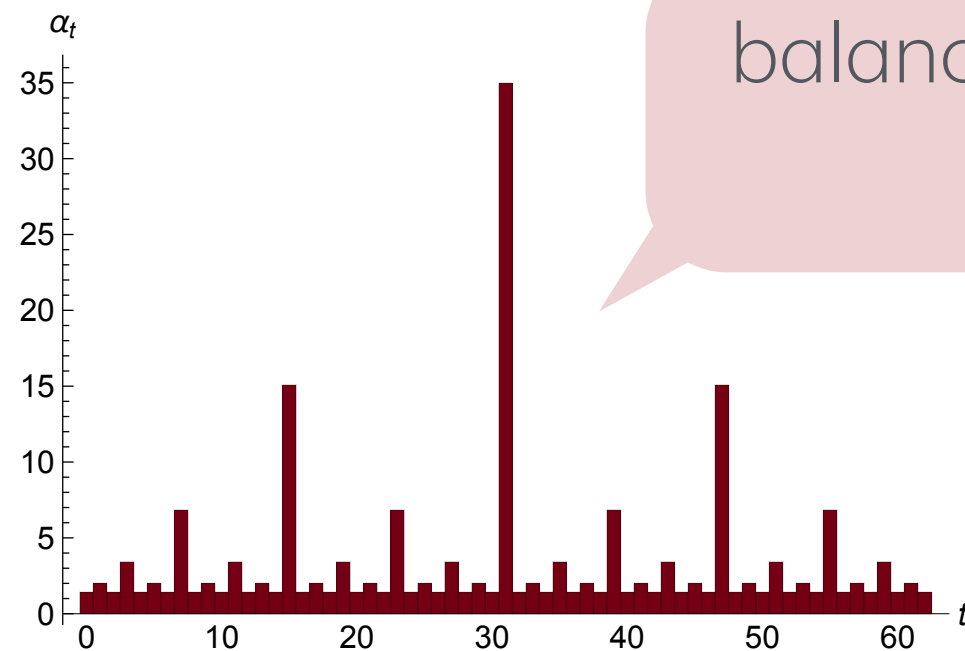
(3/3) Stepsize diagram



Related: long steps

Theorem. Let L be convex and smooth. For GD with *silver* stepsize scheduler $(\alpha_s)_{s \geq 0}$ and $t = 2^k - 1$, we have

$$L(\theta_t) - \min L = O(1/t^{1.27})$$



balance performance in high/low curvatures:
 $0.5\theta^2$ vs Huber function

- cover more problems, e.g., quadratics
- less practical stepsize scheduler

Altschuler, Parrilo. “Acceleration by stepsize hedging II: silver stepsize schedule for smooth convex optimization.” Mathematical Programming 2024

Grimmer, Shu, Wang. “Composing optimized stepsize schedules for gradient descent.” Mathematics of Operations Research 2025

Summary

training instability caused by large stepsize

acceleration via large stepsize: three ML examples

new mental picture: valley

general
losses

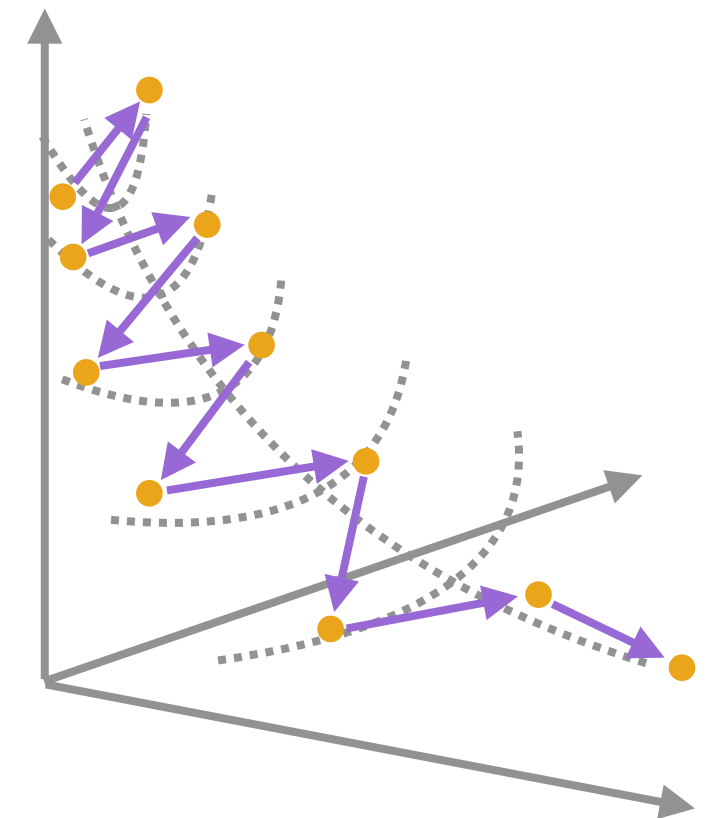
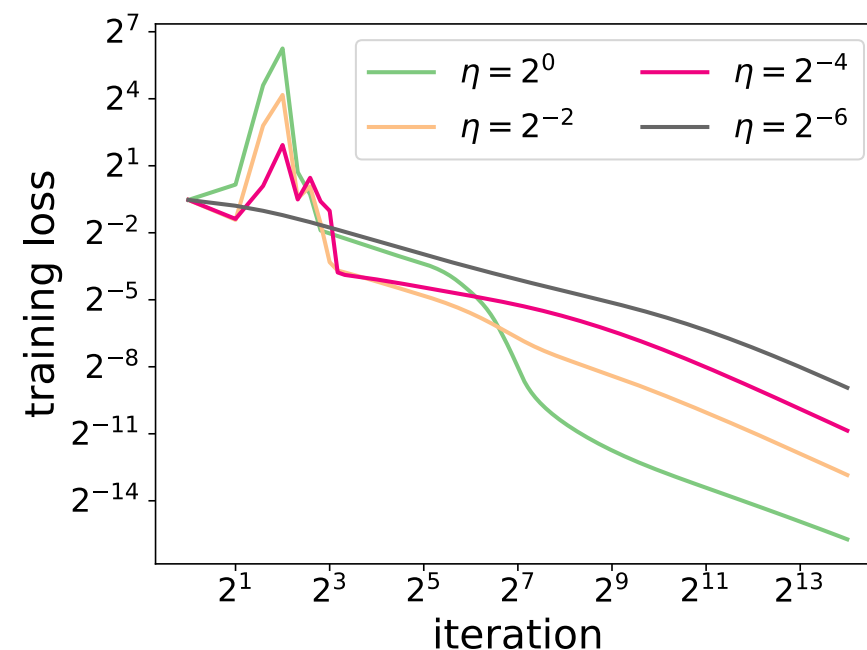
neural
networks

implicit bias,
generalization

cross entropy, attention, ...

practice

theory



Open problems (set 1/2)

Call for clear, rigorous understanding on

🤔 what functional property enables large stepsize?

🤔 trackable measures of trajectory: sharpness? local mean?

🤔 early-phase feature learning, especially against NTK?

🤔 large stepsize for other optimizers, e.g., SGD, Adam?

Open problems (set 2/2)

Call for useful, heuristic insights on

- 💡 better stepsize schedulers, e.g., warmup, stepsize decaying?
- 💡 better optimizer, e.g., preconditioning, normalization?
- 💡 interplay between stepsize vs structure, e.g., attention, depth?
- 💡 how to understand other instabilities, e.g., data, precision?

Q & A

